

# STRUCTURED QUERY LANGUAGE(SQL)

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# Database Language

- Database languages are specialised programming languages used to interact with a Database Management System (DBMS) to define, manipulate, control, and manage data.
- The most common language for relational databases is SQL (Structured Query Language), which incorporates four main types of database languages.
  - 1.*DDL (Data Definition Language)*
  - 2.*DCL (Data Control Language)*
  - 3.*DML (Data Manipulation Language)*
  - 4.*TCL (Transaction Control)*

# SQL

- Structured Query Language (SQL) is the standard language used to interact with relational databases.
- Mainly used to manage data: whether you want to create, delete, update or read data, SQL provides commands to perform these operations.
- Widely supported across various database systems like **MySQL, Oracle, PostgreSQL, SQL Server and many others.**
- Mainly works with **Relational Databases** (data is stored in the form of tables)

# History of SQL

- SQL was invented in the 1970s based on the relational data model. *It was initially known as the structured English query language (SEQUEL).*
- The term was later shortened to SQL.
- Oracle, formerly known as Relational Software, became the first vendor to offer a commercial SQL relational database management system.

# SQL Commands

- Structured query language (SQL) commands are specific keywords or SQL statements that developers use to manipulate the data stored in a relational database.

## Data definition language

- Data definition language (DDL) refers to SQL commands that design the database structure.
- Database engineers use DDL to create and modify database objects based on the business requirements. For example, the database engineer uses the CREATE command to create database objects such as tables, views, and indexes.

## Data query language

- Data query language (DQL) consists of instructions for retrieving data stored in relational databases. Software applications use the SELECT command to filter and return specific results from a SQL table.

## Data manipulation language

- Data manipulation language (DML) statements write new information or modify existing records in a relational database.
- For example, an application uses the INSERT command to store a new record in the database.

# SQL Commands

## Data control language

- Database administrators use data control language (DCL) to manage or authorize database access for other users.
- For example, they can use the GRANT command to permit certain applications to manipulate one or more tables.

## Transaction control language

- The relational engine uses transaction control language (TCL) to automatically make database changes.
- For example, the database uses the ROLLBACK command to undo an erroneous transaction.

# SQL standards

- SQL standards are a set of formally defined guidelines of the structured query language (SQL).
- The American National Standards Institute (ANSI) and International Organization for Standardization (ISO) adopted the SQL standards in 1986.
- Software vendors use the ANSI SQL standards to build SQL database software for developers.

## SQL injection

- SQL injection is a cyberattack that involves tricking the database with SQL queries. Hackers use SQL injection to retrieve, modify, or corrupt data in a SQL database.
- For example, **they might fill in a SQL query instead of a person's name in a submission form to carry out a SQL injection attack.**

# Data Definition Language(DDL)

- **DDL** stands for **Data Definition Language**. It is used to define database structure or pattern.
- It is used to create schema, tables, indexes, constraints, etc. in the database.
- Using the DDL statements, you can create the skeleton of the database.
- Data definition language is used to store the information of metadata like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.



# Data Definition Language (DDL)

- **Create:** It is used to create objects in the database.
- **Alter:** It is used to alter the structure of the database.
- **Drop:** It is used to delete objects from the database.
- **Truncate:** It is used to remove all records from a table.
- **Rename:** It is used to rename an object.
- **Comment:** It is used to comment on the data dictionary.

# Data Manipulation Language (DML)

- **DML** stands for **Data Manipulation Language**. It is used for accessing and manipulating data in a database. It handles user requests.
- **Select**: It is used to retrieve data from a database.
- **Insert**: It is used to insert data into a table.
- **Update**: It is used to update existing data within a table.
- **Delete**: It is used to delete all records from a table.
- **Merge**: It performs UPSERT operation, i.e., insert or update operations.
- **Call**: It is used to call a structured query language or a Java subprogram.
- **Explain Plan**: It has the parameter of explaining data.
- **Lock Table**: It controls concurrency.

# Data Control Language (DCL)

- It is used to control the access permissions of users to the database.
- DCL commands help grant or revoke privileges to users, determining who can perform actions like reading or modifying data.

**The two main DCL commands are:**

- **Grant:** Gives user access to the database
- **Revoke:** Removes access or permissions

# Grant Command

The GRANT command in a Database Management System (DBMS) is used to provide specific permissions or privileges to users or roles. This command allows administrators to control access to database objects, ensuring that only authorized users can perform certain actions, such as selecting, inserting, updating, or deleting data.

Syntax:

*GRANT privileges  
ON object  
TO user\_or\_role [WITH GRANT OPTION];*

*Example:*

*GRANT SELECT, INSERT ON students TO user;*

# Revoke Command

The **REVOKE** command in a Database Management System (DBMS) is used to remove previously granted permissions or privileges from users or roles. This command is essential for **managing access control**, ensuring that users do not have more privileges than necessary to perform their tasks.

*Syntax:*

***REVOKE privileges ON object FROM user\_or\_role;***

*Example:*

***REVOKE ALL PRIVILEGES ON students FROM user;***

# Transaction Control Language (TCL)

- The TCL full form is are used to manage and control transactions in a database, grouping them into logical units. These commands help ensure the integrity of data and consistency during complex operations.
- Here are the two main commands in this category:
  - **Commit:** It is used to save the transaction on the database.
  - **Rollback:** It is used to restore the database to original since the last Commit.

# COMMIT

- The COMMIT command is used to save all changes made during a transaction in the database. This command ensures that modifications made by DML statements (such as INSERT, UPDATE, or DELETE) become permanent in the database.

- Syntax

Database Operation

Commit

```
mysql> INSERT INTO ClassMembers (StudentID, Name, Age, Weight)
      -> VALUES (4, 'Emma Johnson', 23, 50);
Query OK, 1 row affected (0.03 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

# ROLLBACK

- The rollback command is used to restore the database to its state at the last COMMIT, effectively undoing any changes made since that point. It helps ensure data consistency by allowing the reversal of partial or erroneous operations.

## Syntax

- ROLLBACK;

```
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> commit;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> DELETE FROM ClassMembers where StudentID = 4;  
Query OK, 1 row affected (0.03 sec)  
  
mysql> rollback;  
Query OK, 0 rows affected (0.00 sec)
```



# SQL DATATYPES

| Category  | Data Type    | Use                  | Example Syntax          | Brief Explanation                                        |
|-----------|--------------|----------------------|-------------------------|----------------------------------------------------------|
| NUMERIC   | INT          | Whole numbers        | Age INT                 | Stores integers without decimal                          |
|           | DECIMAL(p,s) | Exact decimal values | Salary<br>DECIMAL(10,2) | <b>p</b> = total digits, <b>s</b> = digits after decimal |
|           | FLOAT        | Approximate decimals | Marks FLOAT             | Used for scientific/approx values                        |
| CHARACTER | CHAR(n)      | Fixed-length text    | Gender CHAR(1)          | Always occupies fixed space                              |
|           | VARCHAR(n)   | Variable-length text | Name VARCHAR(50)        | Uses space as per data length                            |
|           | TEXT         | Long text            | Address TEXT            | Stores large text data                                   |
| DATE/TIME | DATE         | Date only            | DOB DATE                | Format: YYYY-MM-DD                                       |
|           | TIME         | Time only            | LoginTime TIME          | Format: HH:MM:SS                                         |
|           | DATETIME     | Date + Time          | CreatedAt DATETIME      | Stores both date and time                                |
| BOOLEAN   | BOOLEAN      | True / False         | IsActive BOOLEAN        | Stores logical values                                    |